Individual Differences in Neural Oscillations and Behavioural Efficiency During Cognitive Conflict: An EEG Study

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Abstract

This study investigated whether individual differences in neural oscillatory responses predict behavioural efficiency during cognitive conflict. EEG and ECG were recorded in fifty-four healthy young adults during a Stroop task. The final sample sizes were N = 51 for behavioural analysis, N = 52 for neural and physiological analysis, and N = 48 for regression analysis. Behavioural efficiency was measured by the Inverse Efficiency Score (IES). Reaction times were significantly slower, accuracy was significantly lower, and IES was higher for incongruent trials compared to congruent trials, indicating a Stroop interference effect. Theta power and heart rate demonstrated significant increases, whereas alpha power was significantly suppressed, compared to resting baseline. Multiple regression identified theta power increase as the only significant individual predictor of IES *(B = 75.03, p = .007)*. Findings suggest that theta power may serve as a marker of compensatory cognitive effort rather than neural efficiency, with implications for EEG-based assessment of cognitive control.

Introduction

Executive functions such as managing thoughts, actions, and feelings are fundamental to traverse everyday life cognitive complexity, enabling individuals to prioritise tasks, resist impulses, and sustain concentration when faced with distractions (Miyake et al., 2000). When combined, all the actions mentioned fall into the category of cognitive control; a multi-mechanism function responsible for regulating thoughts and actions, to facilitate goal-directed behaviour (Badre, 2024; Miller & Cohen, 2001). Cognitive control includes mechanisms that are recruited dynamically depending on task demands, meaning they are variable and adaptive. The relevant mechanisms of the present study are selective attention, inhibitory control and conflict monitoring (Duncan, 2013; Logan, 2017). Conflict monitoring is the brain's capacity to identify challenging simultaneous responses, whilst signalling the necessity for increased cognitive control according to the task at hand (Botvinick et al., 2001). The conflict theory suggests that post-detection neural oscillations adapt to provide greater cognitive control and attentional bias toward task-relevant information, facilitating the adjustment of an adequate response (Botvinick et al., 2001). This flexible alteration of control helps sustain goal-directed behaviour even when irrelevant information competes with the processing priority (Miller & Cohen, 2001; Norman & Shallice, 1986). Impaired cognitive control is linked to conditions such as ADHD, schizophrenia, and autism, as deficits in goal-directed thought and response inhibition can lead to impulsive and inattentive behaviours (Riglin et al., 2022). Despite the essential function of cognitive control, neural and physiological mechanisms underlying effective control remain poorly understood, especially concerning how these mechanisms vary across individuals (Michon et al., 2025). The Stroop task is a widely used paradigm for studying cognitive control, providing a controlled setting in which individuals can resolve cognitive interference while maintaining goal-directed focus (Stroop, 1935). Nonetheless, behavioural results alone are insufficient to explain the diverse neural strategies potentially employed to reach similar outcomes, indicating that neural oscillatory activity could offer insights into behavioural scores (Michon et al., 2025).

In the Stroop task, the participant must identify the ink colour of presented words. Congruent trials show matching words and ink colours (e.g., "RED" in red ink), whereas incongruent trials display mismatches (e.g., "RED" in blue ink) (Stroop, 1935). Incongruent trials usually result in a significant interference effect, as the automatic processing of semantic meaning in the word competes with the target aim of naming the ink colour, characterised by slower reaction times and lower accuracy (Botvinick et al., 2001; Desimone & Duncan, 1995). This

conflict can be explained as automaticity, a theory stating that word reading is the prepotent response, due to being more frequently practised outside the experimental task, requiring less cognitive demand in comparison to ink detection, leading to the interference effect (LaBerge & Samuels, 1974). As a result, the Stroop interference effect is a robust metric for assessing efficiency of conflict detection and resolution, thereby providing a valid method for examining variations in the interference effect across individuals during conflict processing (Ménétré & Laganaro, 2023; Miller & Cohen, 2001). The persisting question is whether individual differences in behavioural performance reflect noticeable changes in neural oscillatory activity patterns during conflict processing.

The neural recording system which best identified the relevant oscillatory markers was electroencephalography (EEG), as it captured occipital alpha and task-related theta with remarkable temporal resolution. Theta power (4-8 Hz) findings indicate that theta power increases occur when the brain signals require greater control, suggesting that theta is a communication mechanism between frontal control regions of executive functioning and conflict monitoring, coordinating task-relevant neural processes to resolve conflict (Cavanagh & Frank, 2014). Alpha power (8-13 Hz), typically recorded over Oz, is linked to attentional gating and cortical inhibition. While rises in alpha power are associated with the suppression of irrelevant brain areas, the decrease in alpha power, empirically known as alpha suppression, signifies increased cortical excitability in sensory regions relevant to the task, indicating heightened attentional focus (Jensen & Mazaheri, 2010). Together, the increase of task-related theta and suppression of alpha (Oz) provide complementary indices of the neural demands associated with conflict processing (Cavanagh & Frank, 2014; Jensen & Mazaheri, 2010). Moreover, evidence has shown a reciprocal synchronisation between theta and alpha oscillations, suggesting the interplay of top-down executive functions, and the engagement of attentional priorities coordination with cognitive resources during demanding tasks (Huycke et al., 2021).

Whilst neural oscillation measures reflect the brain's response to cognitive demands, the physiological responses in the autonomic nervous system, like increased heart rate (BPM), indicate sympathetic activation during cognitive demand. Highlighting the link between heart rate and conflict monitoring, serving as a peripheral physiological indicator of arousal during cognitive effort (Kahneman, 1973). Despite this, the connection between physiological responses and cognitive control remain largely unexplored creating a significant gap in the literature. Therefore, the present study secondarily aimed to investigate physiological

response's role in conflict resolution efficiency, by measuring heart rate alongside EEG during the Stroop task.

A critical question concerns the relationship between the intensity of neural responses and the level of behavioural performance. According to the neural efficiency theory, highly capable cognitive systems are characterised by focused and specific neural activation rather than broad or indiscriminative engagement of resources (Haier et al., 1992; Neubauer & Fink, 2009). However, within this framework, stronger neural signals do not automatically equate to superior task performance; individuals with higher cognitive ability typically achieve equivalent or better outcomes while using less neural effort, by efficiently utilising enhanced neural synchrony, as opposed to increased activation, as precisely timed oscillations generate more robust and effective signals than amplified uncoordinated activity (Neubauer & Fink, 2009). This framework is empirically supported by several methodologies. Research by Neubauer et al. (2002) using EEG to measure upper alpha band desynchronisation during cognitive tasks, demonstrating that higher IQ individuals showed less alpha ERD, indicating more precise neural activation. Grabner et al., (2006) elaborated this further using EEG, testing tournament chess players on tasks of mental speed, memory and reasoning, demonstrating that neural efficiency extends beyond general intelligence to domain-specific expertise. Additionally, research by Wartenburger et al. (2009) indicated a decrease in functional MRI corresponding with improved task accuracy, with the reduction effect being more pronounced in higher skilled individuals. The difference between allocation and efficiency holds theoretical significance, as excessive neural activation for average performance could indicate compensatory over-activation rather than efficient regulation (Braver, 2012). This suggests that optimal neural efficiency involves calibrating neural effort to task demands, rather than maximising activation. This distinction holds significant relevance to the present study, as individual variations in theta and alpha modulations during conflict processing could indicate differences in neural efficiency, where some individuals may exercise greater neural activations to achieve equivalent behavioural results, while others may resolve conflict through more refined and focused oscillatory responses.

At a behavioural level, combined speed and accuracy best reflect performance on conflict tasks. Effective performance in conflict-related tasks like the Stroop task requires not only fast responses or high accuracy but an optimal calibration between the two. The Inverse Efficiency Score (IES) quantifies this relationship by dividing mean reaction time by the proportion of correct responses; the resulting single metric effectively penalises both delayed

responses and high error rates (Snodgrass et al., 1985); lower IES indicates more efficient performance. Examining IES instead of two separate metrics (Reaction times and accuracy) offers a better indicator of conflict resolution effectiveness and behavioural index for neural modulation.

There is sufficient group-level research demonstrating that theta power and alpha suppression are related to cognitive conflict, however, the individual variations factoring behavioural efficiency and physiological responses, have not been further explored (Hanslmayr et al., 2008). As research has indicated associations between cardiac autonomic activity and cognitive control performance, including executive function, response inhibition, and working memory, but have yet to explain task-induced heart rate variations during conflict processing in relation to individual behavioural efficiency (Forte & Casagrande, 2024; Riganello et al., 2023). Past studies established reliable group effects; however, few studies have directly investigated the link between greater neural modulation during conflict tasks and effective performance (IES). The key theoretical and practical question is whether greater neural efficiency predicts greater behavioural efficiency. Resolving this would theoretically clarify whether oscillatory modulation indicates effective cognitive control or compensatory cognitive load, a difference that has substantial implications for understanding EEG performance markers. Furthermore, discerning neural and physiological indicators of conflict resolution could practically enhance cognitive function assessment in clinical populations exhibiting deficits in cognitive control, such as those diagnosed with ADHD or schizophrenia.

Therefore, the present study examined whether individual differences in theta and alpha modulation during the Stroop task correlated with behavioural efficiency, measured by IES. Based on past literature, three directional hypotheses were proposed: (1) Stroop behavioural performance: incongruent trials will produce significantly slower reaction times and lower accuracy, producing greater IES than congruent trials, (2) Task-related neural and physiological responses: cognitive tasks will elicit significantly greater theta power, significantly suppressed alpha power, and elevated heart rate in the Stroop task in relation to baseline, (3) Given that theta, alpha, and heart rate provide complementary indicators of neural and physiological demand, individual variability in task-induced theta increase, alpha decrease, and heart rate change during the Stroop task will significantly predict behavioural efficiency measured by IES.

Methods

Participants

The participants included fifty-four healthy young adults (19 male and 35 females, range = 18-25, *M = 19.9*, *SD = 1.2*), recruited via convenience sampling, online participant database (Aston University SONA), and personal networks via word of mouth. Before the recruitment process, a priori G-power analysis was conducted, revealing the minimum sample requirement of 34 participants (N = 34) to achieve 80% power in detecting a moderate effect size *(d = 0.5)* at a significant level of $\alpha = .05$. The moderate effect size *(d = 0.5)* had been chosen due to consistent reporting in previous EEG studies associated with cognitive control, as a reliable indicator of the conflict-related oscillatory effects (Cavanagh & Frank, 2014; Hanslmayr et al., 2008). Of the 54 participants, two were excluded before calculations: one for duplicating participation (registered twice), and one due to absence of complete data. Additionally, post-extractions, one participant was excluded from behavioural analysis due to 0% accuracy in the Stroop task, and four participants were excluded from regression analysis due to not having complete change scores for regression variables. Therefore, the final analysis sample consisted of N = 51 for Stroop behavioural analysis, N = 52 for neural and physiological analysis (as participant excluded from behavioural analysis still displayed valid neural and physiological data), and N = 48 for regression analysis. Inclusion criteria required normal or corrected vision and normal colour vision, while exclusion criteria involved any neurological disorders, epilepsy, and psychoactive medication. The study was conducted under a low-risk umbrella ethics framework, approved by the Aston Psychology Research Ethics Committee, and informed consent was obtained by all participants.

The age demographic of 18 to 25 years was selected to increase internal validity by minimising age-related variations in reaction times (RTs) and cognitive processes (Constantinidis & Luna, 2019; Dykiert et al., 2012; Forte et al., 2024; Ménétré & Laganaro, 2023).

Design and task

The study utilised a within-subjects design to assess changes in neural and physiological measures between task and baseline. Additionally, correlational analyses were used to examine whether individual variations in psychophysiological indicators predicted behavioural efficiency. The Stroop task trials consisted of congruent (word ‘RED’ in red ink) and incongruent conditions (word ‘RED’ in blue ink). Participants were instructed to respond

to ink colour while ignoring word meaning using keyboard arrow keys: left arrow for yellow ink, up arrow for red ink, and right arrow for blue ink. Each trial began with a 250 ms fixation cross, followed by the presentation of the word which remained on screen until a response was made; post response, a 400 ms feedback period would produce a red screen and unpleasant honk sound for incorrect responses or a pleasant bell auditory tone if correct. After a block of 9 practice trials, participants completed 8 blocks of 18 trials each containing 6 congruent and 12 incongruent trials, with a 10 second inter-block fixation rest period.

Prior to the Stroop task, participants had to complete a 3-minute baseline alternating between 30 seconds of eyes-open (EO) and 30 seconds of eyes-closed (EC) to establish resting neural and physiological baseline values to compare against task-induced changes.

Subsequently, participants also underwent a Stop-signal task and a PASAT task, as part of data collection for a wider project, however, these will not be elaborated further, as these tasks are not included in this research report.

Behaviour Acquisition

Tasks were delivered using PsychoPy (version 2024.2.4).

Trials with RTs below 200 ms or above 1200 ms were excluded from analysis.

Behavioural performance was quantified using reaction time (RT) and accuracy. Behavioural efficiency was calculated using the Inverse Efficiency Score (IES = Mean RT / Proportion Correct), where higher scores indicate lower efficiency (Sciaraffa et al., 2021).

IES was chosen over the individual measurement of raw reaction time and accuracy to moderate the speed-accuracy trade-off, nullifying the prioritisation of one metric in participants' outcomes, thereby evening disparities in results.

Neural Oscillatory Acquisition and Analysis

EEG was recorded using one occipital electrode at Oz, using the 10-20 convention at 10% of the nasion-to-inion distance from inion, one ground electrode at the top left corner of the forehead and one reference electrode positioned at Fz. Raw EEG data (0.5-35 Hz) was recorded using a BIOPAC MP36 system (BIOPAC Student's Lab 4.0) keeping impedances below 20 kΩ, with a Stimtracker connected via USB transmitting stimulus onset triggers from PsychoPy, ensuring temporal synchronisation between task events and neural data. Raw EEG was bandpass filtered into alpha (8-13 Hz) and theta (4-8 Hz) frequency bands and

power for each was quantified as the Root Mean Square (RMS) of the respective filtered signal. Theta power was extracted during the 200-600 ms post-stimulus window of the Stroop task and compared against that of the inter-block 10 second baseline interval.

ECG was simultaneously being recorded via two electrodes placed on the back of the left hand and one electrode on the pulse point of the left side of the neck. Heartbeat intervals (HBI) were detected automatically via BIOPAC QRS peak detection and were converted to beats per minute (BPM). Mean heart rate was extracted across task and inter-block baseline periods.

Procedure and Statistical Analysis

All statistical analyses were conducted using Jamovi (version 2.6.44). Paired-samples t-tests were conducted to assess task versus baseline differences in alpha power, theta power, heart rate and Stroop task behaviour. Multiple regression analyses were conducted to determine whether alpha power, theta power and heart rate predicted behavioural performance (IES) on Stroop.

Spearman correlations, which are non-parametric, were employed for variables that did not meet normality assumptions. For within-subject comparisons, paired-samples t-tests were retained. Because the central limit theorem indicates that these tests are generally robust to moderate deviations in normality, particularly in similar sample sizes (N = 30) (Field, 2018).

Results

1. Stroop Task Behavioural Performance

As shown in Table 1, accuracy of congruent trials was significantly higher *(M = 96.6%, SD = 4.1)* in comparison to incongruent trials *(M = 95.4%, SD = 3.9), t(50) = 2.66, p = .010, d = 0.37*. Similarly, reaction times were significantly faster for congruent trials *(M = 628 ms, SD = 142)* compared to incongruent trials *(M = 685 ms, SD = 154), t(50) = -6.13, p < .001, d = 0.86.* Incongruent trials generated significantly higher IES (*M = 719 ms, SD = 167)* than congruent trials *(M = 648 ms, SD = 138), t(50) = -6.02, p < .001, d = 0.84*, confirming Hypothesis 1 of the Stroop interference effect.

Table 1. Performance metrics of the Stroop task (N = 51)

Measure	Congruent M (SD)	Incongruent M (SD)	*t*(50)	*p*	*d*
Accuracy (%)	96.6 (4.1)	95.4 (3.9)	2.66	.010	0.37
Mean RT (ms)	628 (142)	685 (154)	-6.13	< .001	0.86
IES (ms)	648 (138)	719 (167)	-6.02	< .001	0.84

Note: *IES = Inverse Efficiency Score, and d = Cohen's d*

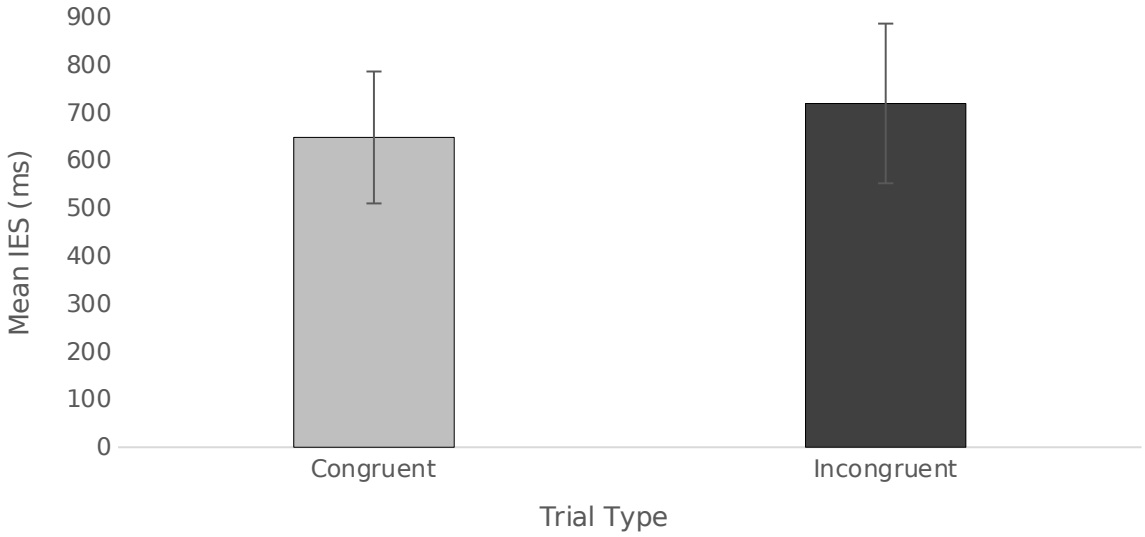


Figure 1. Mean Inverse Efficiency Scores (ms) for congruent and incongruent Stroop trials (N = 51). Error bars represent ±1 standard deviation.

2. Task-Related Neural and Physiological Responses

2.1 Stroop Task Theta Power

Theta power (4-8 Hz) (N = 52) displayed a significant increase during the Stroop task *(M = 5.17, SD = 2.19)* in comparison to the baseline interval *(M = 4.15, SD = 1.95), t(51) = -7.96, p < .001, d = 1.10,* consistent with Hypothesis 2 of greater theta power during conflict-processing.

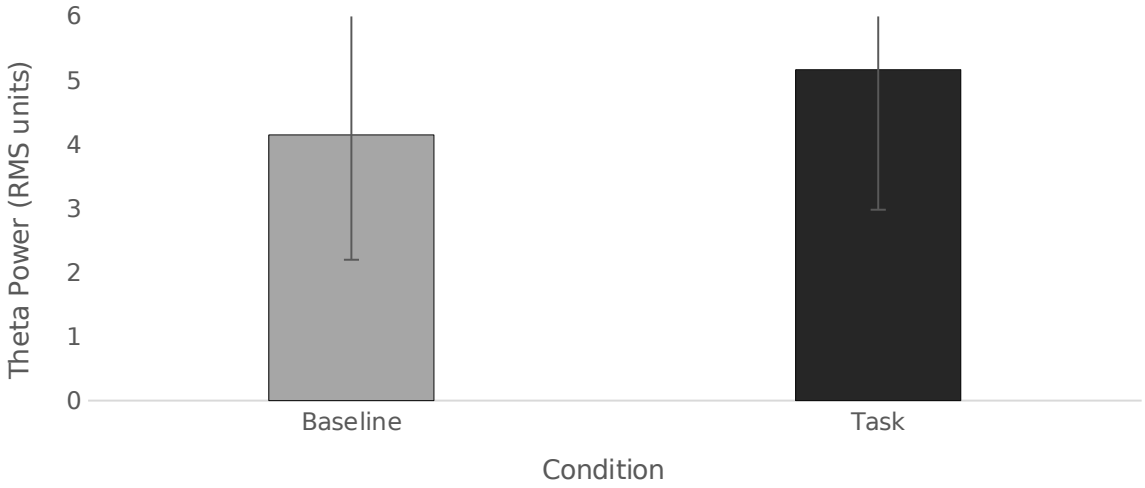


Figure 2. Mean theta power (RMS units), during baseline and Stroop task conditions (N = 52). Error bars represent ±1 standard deviation.

2.2 Stroop Task Alpha Power: Task and Baseline

Alpha power (8-13 Hz) was compared between eyes-open (EO) and eyes-closed (EC) resting conditions. EO alpha power *(M = 3.63, SD = 1.51)* was significantly lower than EC alpha power *(M = 6.08, SD = 2.68), t(51) = 8.43, p < .001, d = 1.17.* As shown in Table 2, alpha power was significantly lower during the Stroop task *(M = 3.81, SD = 1.26)* compared to baseline *(M = 4.02, SD = 1.44), t(51) = 2.50, p = .016, d = 0.35,* consistent with hypothesis 2 predicting alpha suppression during conflict resolution.

Table 2. Alpha power (RMS units): Task vs. Baseline (N = 52)

Task	Baseline M (SD)	Task M (SD)	Change M (SD)	*t*(51)	*p*	*d*
Stroop	4.02 (1.44)	3.81 (1.26)	-0.20 (0.58)	2.50	.016	0.35

Note: *negative change expresses suppression (baseline > task), d = Cohen's d*

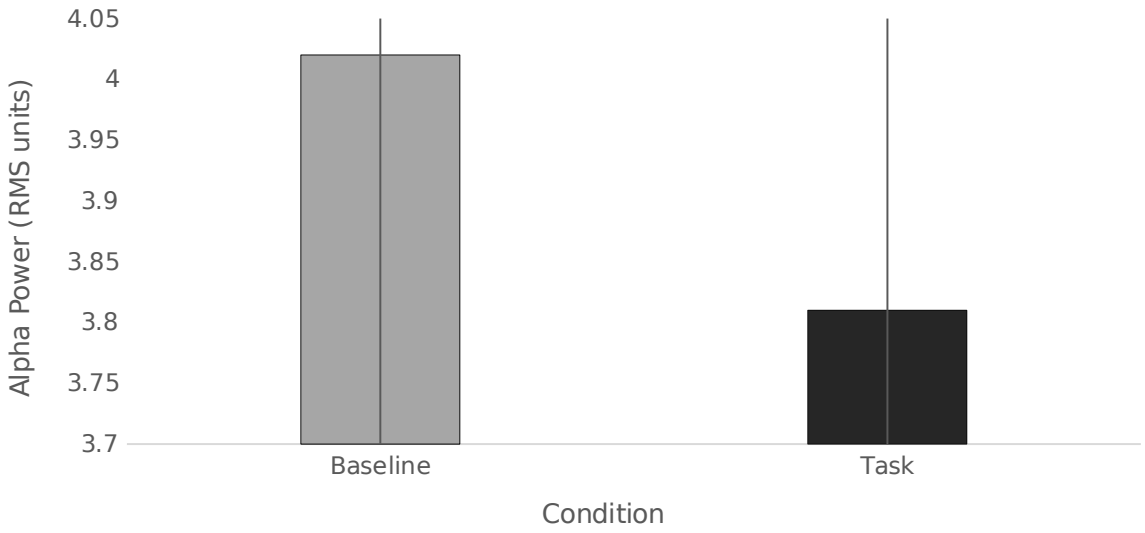


Figure 3. Mean alpha power (RMS units), during baseline and Stroop task conditions (N = 52). Error bars represent ±1 standard deviation.

2.3 Heart Rate: Stroop Task and Baseline

Mean resting heart rate during eyes-open was 67.7 BPM *(SD = 7.0)* and 68.0 BPM *(SD = 4.4)* during eyes-closed conditions. Additionally, heart rate was significantly higher during the Stroop task *(M= 69.5, SD = 6.6)* compared to baseline *(M = 66.4, SD = 7.2), t(51) = -4.85, p < .001, d = 0.67,* as shown in Table 3, consistent with hypothesis 2 of cognitive conflict eliciting physiological responses.

Table 3. Heart rate (BPM): task and baseline (N = 52)

Task	**Baseline M (SD)**	**Task M (SD)**	**Change M**	***t*(51)**	***p***	***d***
Stroop	66.4 (7.2)	69.5 (6.6)	+3.1	-4.85	< .001	0.67

Note: *d = Cohen's d*

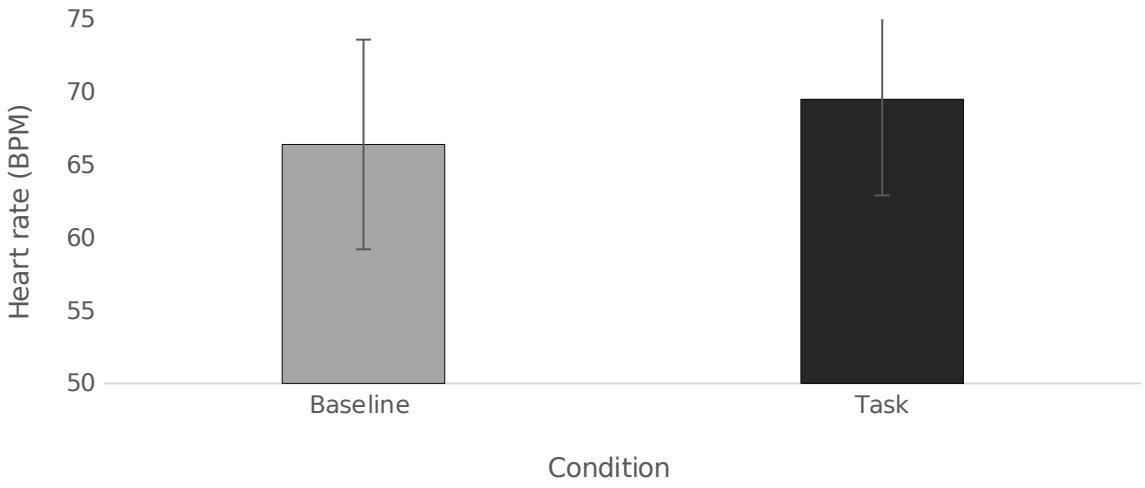


Figure 4. Mean heart rate (BPM), during baseline and Stroop task conditions (N = 52). Error bars represent ±1 standard deviation.

3. Multiple Regression: Predictors of Individual Behavioural Efficiency

Shapiro-Wilk tests indicated that the normality assumption was not met by several variables *($p < .05$)*, including accuracy *($W = 0.834$, $p < .001$)*, reaction times *($W = 0.929$, $p = .006$)*, IES *($W = 0.905$, $p < .001$)*, and alpha change scores *($W = 0.861$, $p < .001$)*. The only variable which met normality assumptions was theta power change *($W = 0.975$, $p = .399$)*, however, as the majority of variables violated normality, Spearman's rho correlation replaced Pearson correlations to explore the associations among neural, physiological and behavioural variables.

Spearman rank correlations (Table 4) and multiple regression analyses (Table 5) were conducted to examine whether individual differences in alpha suppression, theta power increase, and heart rate change during the Stroop task predicted behavioural efficiency (IES; N = 48).

As shown in Table 4, Spearman rank correlations revealed that theta power increase showed a significant positive correlation with incongruent IES *($\rho = .360$, $p = .010$)* and congruent IES *($\rho = .317$, $p = .024$)*. A significant inverse correlation was found between theta increase and alpha suppression *($\rho = -.280$, $p = .045$)*. Alpha suppression and heart rate changes were not significantly correlated with IES *($p > .46$)*.

Table 4. Spearman Rho Correlations: Neural, Physiological, and Behavioural (N = 48)

	Theta change (*p*)	Alpha change *(p)*	HR change *(p)*	Incongruent IES *(p)*	Congruent IES *(p)*
Theta change	_____				
Alpha change	-.280 (.045)	_____			
HR change	-.134 (.344)	-.221 (.115)	_____		
Incongruent IES	.360* (.010)	-.105 (.461)	.045 (.752)	_____	
Congruent IES	.317* (.024)	-.075 (.598)	-.052 (.717)	.926*** (<.001)	_____

Note: **p < .05, ***p < .001; Values represent Spearman's ρ*

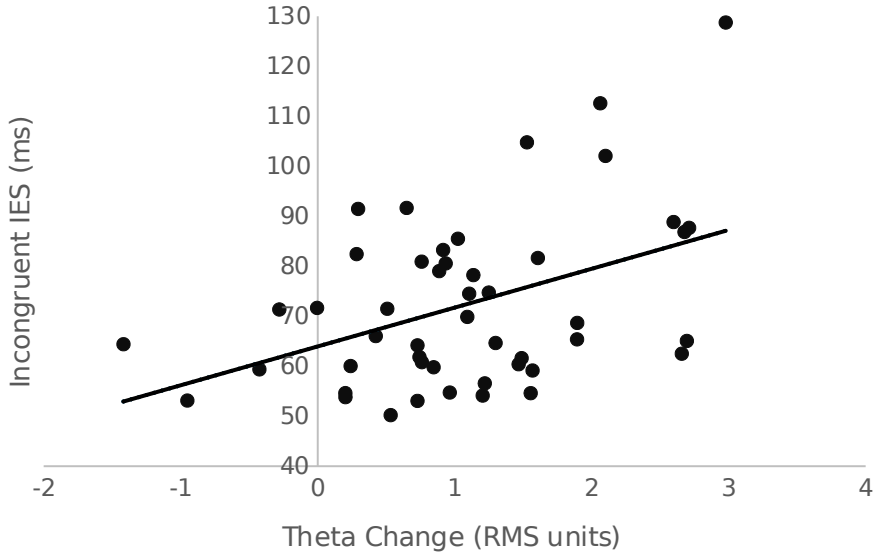


Figure 5. Scatter plot relationship between theta power change and incongruent IES (N = 48).

Presented in Table 5, the regression model for incongruent IES was significant, *F(3, 44) = 3.55, p = .022, R^2 = .195, adjusted R^2 = .140.* Theta power increase was the only significant individual predictor *(B = 75.03, t(44) = 2.86, p = .007)*. Alpha suppression *(B = 0.30, t(44) = 0.01, p = .991)* and heart rate change *(B = 15.62, t(44) = 0.66, p = .510)* were not significant. The model predicting congruent IES was also significant, *F(3, 44) = 2.83, p = .049, R^2*

= .162, with theta being the only significant predictor again *(B = 52.79, t(44) = 2.39, p = .021).*

Variance inflation factors (VIF) confirmed no multicollinearity among predictors *(theta VIF = 1.33, alpha VIF = 1.32, heart rate VIF = 1.07)*, supporting the independence of regression predictors. Standardised theta coefficients indicated theta power *(β = .446)* as the strongest predictor, in comparison to alpha *(β = .002)* and heart rate *(β = .093)*, confirming theta as the dominant individual predictor of behavioural efficiency. Partially supporting hypothesis 3, as theta predicted behavioural efficiency while alpha and heart rate did not.

Table 5. Multiple Regression: Predictors of incongruent IES (N = 48)

Predictor	***B***	**SE**	***t*(44)**	***p***	**Sig.**	***β***
Alpha suppression	0.30	26.20	0.01	.991	ns	.002
Theta increase	75.03	26.28	2.86	.007	**	.446
Heart rate change	15.62	23.51	0.66	.510	ns	.093

Note: *B = unstandardised regression coefficients; R^2 = .195, Adjusted R^2 = .140, F(3, 44) = 3.55, p = .022; ** p < .01.*

Discussion

Behavioural analysis confirmed the Stroop interference effect, as the incongruent trials displayed significantly slower reaction times and lower behavioural efficiency compared to congruent trials. Theta power elevated significantly *(d = 1.10)*, while alpha power was significantly suppressed *(d = 0.35)*, both in relation to their respective baselines. Heart rate also increased during the Stroop task in relation to the resting measure. Regression analyses demonstrated that theta power amplitude significantly predicted poorer behavioural efficiency, whereas alpha suppression and heart rate did not.

The Stroop interference effect demonstrated a large effect size *(d = 0.86)*, with a 57 milliseconds (ms) difference in reaction times (RT). As hypothesised (shown in H1), incongruent trials demonstrated slower reaction times, lower accuracy and higher Inverse Efficiency Scores (IES) in comparison to congruent trials (Botvinick et al., 2001; Ménétré & Laganaro, 2023). LaBerge and Samuels (1974) explain this as automaticity in word reading, suggesting that highly practised tasks such as semantic reading dominate over the task goal of identifying ink colours, as reading is more frequently used outside the experimental task. The use of IES further supported this effect, by combining both dependent variables of speed and accuracy into a single metric, allowing a more thorough analysis of performance. The replication of a successful behavioural manipulation task supported by a large interference effect is crucial to present the validity and reliability of the Stroop task as a measure of cognitive conflict, essential to examine related neural and physiological responses.

Neural oscillatory marker theta power increased in relation to baseline and produced a significant effect size *(d = 1.10)*. This supports hypothesis (H2), by showing that theta power increase is consistent with its established role as a mechanism for top-down cognitive control while supporting theta amplitude as a representation of executive control increase during conflict resolution (Cavanagh & Frank, 2014; Hanslmayr et al., 2008). Nevertheless, the interpretation of the theta effect size should be treated with caution, as the EEG system did not include frontal electrode sites (Fz) typically associated with conflict monitoring and executive control in the ACC and PFC neural regions (Cavanagh & Frank, 2014). On the other hand, recent research from Haciahmet et al. (2022) does counter this, showing that Stroop conflict detection can involve posterior theta activity (at Oz) while also predicting conflict errors. As theta was extracted 200-600 ms post-stimulus, these results likely reflect reactive control during conflict detection and resolution, rather than preparation occurring before the stimulus (Braver, 2012).

Alpha suppression was significant, producing lower alpha during the task in comparison to baseline, only generating a minor effect size *(d = 0.35)*, but also aligning with hypothesis 2. Alpha suppression reflected existing literature on blocking irrelevant neural processes (attentional gating) and the increased cortical excitability associated with alpha decreases, suggesting that attentional resources were appropriately localised during the Stroop task, particularly in harder trials (incongruent) (Desimone & Duncan, 1995; Hanslmayr et al., 2008; Huycke et al., 2021). Notably, alpha was extracted from the single occipital electrode (Oz) which is the standard site procedure of measuring alpha fluctuations, making this a methodological strength (Jensen & Mazaheri, 2010). The resting baseline demonstrated this clearly, showing that the eyes-closed condition had greater alpha power than the eyes-open condition, as the visual cortex is inhibited when there is no visual input to process (Jensen & Mazaheri, 2010); validating the EEG system signal quality for capturing genuine alpha oscillatory activity prior to conducting any analysis of alpha. However, the effect size for alpha *(d = 0.35)* was considerably smaller than theta *(d = 1.10)*, suggesting that alpha was a less substantial neural response to conflict. This effect size deviation may be reflective of data extraction differences, as alpha was averaged across the full task period, including congruent and incongruent trials which may have diluted the conflict-specific suppression which occurs during harder incongruent trials, whereas theta was solely extracted from the conflict-relevant post-stimulus window (200-600 ms) (Jensen & Mazaheri, 2010). To better assess conflict-related alpha modulation and individual behavioural efficiency, future research should focus on extracting alpha power from conflict-relevant incongruent trials, rather than averaging across the full task period.

Heart rate increased during the Stroop task compared to baseline, producing a moderate effect size *(d = 0.67)*, consistent with hypothesis 2. This aligns with Kahneman's (1973) theory of sympathetic activation, whereby the sympathetic nervous system raises heart rate in response to increased cognitive demands, indicating that the present Stroop task imposed genuine cognitive demands on participants. The heart rate effect size was intermediate, greater than alpha suppression *(d = 0.35)* but smaller than theta power *(d = 1.10)*, suggesting that heart rate showed a larger physiological reaction to conflict than alpha, while theta remained the dominant indicator of cognitive demand (Cavanagh & Frank, 2014). This cardiac response is consistent with evidence linking autonomic activity to cognitive control performance, implying that the autonomic nervous system was engaged alongside neural oscillations during challenging cognitive tasks (Forte & Casagrande, 2024). The simultaneous recording

of heart rate and EEG provides a meaningful contribution to the largely unexplored literature gap in physiological responses during conflict resolution, demonstrating that involuntary physiological responses occur reliably during conflict resolution and can be measured concurrently with neural data (Forte & Casagrande, 2024; Riganello et al., 2023). However, beats per minute (BPM) is a relatively rudimentary physiological measure, as it reflects broad sympathetic activation, but does not convey the subtle autonomic regulation linked to cognitive control. A more robust physiological metric would be Vagally-mediated HRV, shown to have stronger and more specific associations with cognitive control and executive function (Forte & Casagrande, 2024; Riganello et al., 2023).

The key finding of the study was that theta power increase was the only significant predictor of IES, in a positive direction, as greater theta was associated with worse performance. This suggests that theta reflected the intensity of cognitive load experienced rather than the efficiency of the control process. This pattern is consistent with Braver's (2012) account of compensatory over-activation, whereby individuals who experienced greater difficulty resolving conflict recruited additional neural resources in an attempt to maintain performance, resulting in higher theta power but poorer behavioural efficiency. This is consistent with the neural efficiency theory, which proposed that more capable individuals achieve better performance through more precise and targeted neural activation rather than greater overall activation (Neubauer & Fink, 2009; Haier et al., 1992).

Alpha suppression and heart rate change did not significantly predict IES, suggesting that both measures did not meaningfully differentiate individuals in terms of behavioural efficiency during cognitive conflict; even though both measures demonstrated a reliable group-level response to task demands. The absence of alpha as an individual predictor may be related to the averaging of extraction across the full task, rather than during conflict-specific windows only, such as theta which was only extracted on conflict-relevant trials post-stimulus (200-600 ms). Similarly, the absence of heart rate may be attributed to the lack of specificity required to detect individual differences in cognitive efficiency, using BPM instead of vagally-mediated HRV which could have offered a more sensitive and precise measure of autonomic regulation during cognitive conflict (Forte & Casagrande, 2024; Riganello et al., 2023). The significant inverse correlation between theta increase and alpha suppression *(ρ = -.280, p = .045)* supports the reciprocal relationship between oscillatory measures aligning with evidence of theta-alpha coupling during cognitively demanding tasks (Huycke et al., 2021).

The present study contained several methodological limitations, including the use of a single occipital electrode (Oz), BPM as a physiological measure, and an averaged alpha across the full task period. Additionally, neural efficiency literature typically employs a standardised cognitive ability metric like IQ scores to establish the relationship between theta and IES, not implementing this in the current study may have restricted the comparison of theta to IES (Neubauer & Fink, 2009; Grabner et al., 2006). Future research should incorporate IQ scores to allow direct testing of neural efficiency predictors.

Conclusively, the present study examined whether individual differences in neural oscillatory and physiological responses predicted behavioural efficiency during a Stroop task. The Stroop interference effect was successfully replicated, with theta power, alpha suppression and heart rate all responding significantly to conflict demand. Critically, theta power increase was the only significant predictor of behavioural efficiency, with greater theta being associated with poorer performance rather than better, suggesting that theta reflects cognitive load instead of neural efficiency at the individual level. The results contradicted the interpretation of theta increase as a marker of effective cognitive control while aligning more coherently with the account of compensatory over-activation. Future research should employ frontal-mid electrodes (Fz) and an HRV-based physiological measure, to provide more detailed metrics of the neural efficiency hypothesis in conflict processing paradigms, to better understand cognitive control deficits in clinical populations such as ADHD, schizophrenia and autistic individuals.

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